

Improving Small-Object Detection in CutLER via Multi-Scale Pseudo-Label Generation

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Abstract

Unsupervised object detection has seen remarkable progress through self-supervised visual features, yet reliably detecting small objects without human annotations remains an open challenge. Dominant approaches such as CutLER rely on MaskCut—spectral graph partitioning over frozen DINO features—to generate pseudo-masks for detector training. However, these methods suffer from a severe scale bias: CutLER achieves $AP_l = 32.2$ for large objects but only $AP_s = 1.3$ for small ones on COCO, a gap that fundamentally limits their practical utility. We address this disparity by introducing a DINO feature-contrast heatmap-guided multi-scale crop strategy that selectively re-runs MaskCut at locally distinctive image regions across three crop scales ($0.25\times$, $0.35\times$, $0.50\times$), then merges the resulting masks with the single-scale baseline after filtering and deduplication. Direct pseudo-mask evaluation on 500 COCO validation images shows that our heatmap-guided crop strategy does not improve small-object recall: both baseline and multi-scale pseudo-labels remain near zero at IoU 0.5. This suggests that the selected crops did not produce sufficiently clean, object-centered patch graphs for stable NCut separation. Nevertheless, multi-scale sampling improves medium- and large-object recall by 18% and 16%, respectively, and the combined pseudo-label set lifts bbox AP from 2.14 to 2.26 and more than doubles segmentation AP from 0.48 to 1.08. These findings reframe multi-scale MaskCut as a pseudo-label coverage supplement rather than a small-object rescue mechanism. Overcoming the small-object barrier might require better backbones or fundamentally different grouping strategies beyond crop rescaling. Code is available at <https://github.com/juliavikr/cutler-multiscale>.

1. Introduction

Supervised object detection requires large-scale human-annotated datasets, making it impractical for many domains where annotation is expensive or infeasible. Unsupervised approaches, which learn to detect objects directly from unlabeled images, have recently made substantial progress. CutLER [13] demonstrates that DINO [2] self-supervised features can be used to generate high-quality pseudo-masks via spectral graph partitioning (MaskCut), which are then used to train a Cascade Mask R-CNN detector without any human labels.

Despite this progress, CutLER exhibits a pronounced performance gap between large and small objects. On COCO class-agnostic evaluation, the pre-trained model achieves $AP_l = 32.2$ but only $AP_s = 1.3$ on object detection. This asymmetry is fundamental to the MaskCut design: a ViT-S/8 backbone at 480×480 input maps 8×8 pixel patches, so a COCO small object (area $< 32^2$ px) spans only 3–4 patches. Spectral clustering on such a sparse graph is inherently unstable, and full-image Normalized Cuts tends to favor larger, more coherent regions over small, locally distinctive ones.

A natural hypothesis is that cropping the image around locally distinctive regions and rescaling to the standard input resolution would give MaskCut more patches—and therefore better separability—for small objects. This “zoom-in” intuition has motivated multi-scale and crop-based extensions of self-supervised segmentation methods [4, 9]. We test whether this hypothesis holds for MaskCut-style spectral clustering, and whether it translates into downstream detection improvement.

Our investigation shows that heatmap-guided crop rescaling did not reliably overcome the small-object bottleneck. Although tighter crops can in principle increase the number of ViT tokens covering a small object, this only helps when the crop is well centered on a complete object instance. In practice, many selected crops produced partial

objects, textured regions, or low-salience foreground fragments, leading to unstable NCut partitions and masks that rarely matched COCO small-object instances at sufficient IoU.

Nevertheless, our refined hybrid multi-scale pipeline does improve downstream detection, although not as we expected. Despite our initial objective of improving small object detection and segmentation, only small object segmentation improved, while we observed performance increases for both tasks for medium and large objects. Combined with the baseline single-scale masks, this pseudo-label diversity lifts bbox AP from 2.14 to 2.26 and segmentation AP from 0.48 to 1.08 in our main 5-class run. We believe that our method ended up producing pseudo-labels that cover parts of medium and large objects missed by the single full-image Normalized Cuts pass, while the low-resolution TinyImageNet training images limited how much small-object signal could be recovered during pseudo-label generation.

Contributions.

- We design and implement a DINO feature-contrast heatmap-guided multi-scale MaskCut pipeline that selectively crops and re-segments locally distinctive image regions.
- We conduct a direct pseudo-mask evaluation on 500 COCO validation images, revealing that small-object recall was not improved by our heatmap-guided crop-rescaling strategy, even though medium- and large-object recall increased.
- We demonstrate that combining multi-scale crop masks with baseline single-scale masks improves detector performance through pseudo-label diversity—increased coverage of medium and large objects—rather than direct small-object recovery.
- We provide an ablation study over crop budget and crop scale, identifying the key hyperparameters for balancing recall improvement against mask quality.

2. Related Work

Self-supervised visual features. DINO [2] trains a Vision Transformer (ViT) using self-distillation with no labels. Despite being trained purely on image patches without any annotations, DINO’s attention maps exhibit a strong foreground-background structure, making them well-suited as features for unsupervised object segmentation. Subsequent works such as DINOv2 [7] scaled this approach to larger models and demonstrated further improvements in dense prediction tasks. Our method builds directly on DINO ViT-S/8 features, both for generating pseudo-masks and for computing the feature-contrast heatmap used to guide crop selection.

Unsupervised object segmentation. Normalized Cuts (NCut) [8] is a classic spectral graph partitioning method that finds a balanced bipartition of a similarity graph. TokenCut [15] applied NCut to DINO patch features, demonstrating that the resulting eigenvector cleanly separates foreground from background. LOST [10] localizes objects by finding seeds with low similarity to surrounding patches. Deep Spectral Methods [6] systematically studies the relationship between DINO feature similarity and semantic grouping, showing that multiple eigenvectors correspond to different semantic regions. Our MaskCut baseline follows the same spectral framework, iterating NCut to discover multiple objects per image.

Unsupervised object detection. CutLER [13] combines MaskCut pseudo-mask generation with Cascade Mask R-CNN [1] training and optional self-training, demonstrating competitive class-agnostic detection on COCO without any human labels. SelfMask [9] aggregates multiple saliency maps from different self-supervised models for more robust unsupervised segmentation. DINO-based approaches have been extended to video [14] using temporal correspondence. Our work focuses specifically on the pseudo-label generation stage of CutLER, augmenting MaskCut with multi-scale crop proposals.

Multi-scale and crop-based detection. Image pyramids and multi-scale feature representations are foundational in object detection [5]. Sliding window detectors at multiple scales were standard before deep learning, and Feature Pyramid Networks (FPN) remain central to modern detectors. In the context of self-supervised segmentation, DINO’s multi-crop training strategy uses crops of different sizes to learn scale-invariant features. However, applying multi-scale reasoning at inference time—specifically re-running the segmentation algorithm in image crops—has received less attention. Our work provides an empirical study of this approach in the MaskCut framework, finding that the benefit comes from increased label coverage rather than improved small-object separability.

Small-object detection. Small-object detection remains challenging due to limited resolution and context [3]. Common approaches include super-resolution preprocessing, feature augmentation, and attention mechanisms. In supervised settings, augmenting training data with small object copies (SNIP [11], SNIPER [12]) or using scale-aware feature extractors helps. In the unsupervised setting, the fundamental barrier is that spectral methods require a minimum number of patches to produce stable segmentations. We empirically characterize this barrier for DINO ViT-S/8 at 480×480 input resolution.

3. Method

Our method extends CutLER’s MaskCut pseudo-label generation stage with a DINO feature-contrast heatmap-guided multi-scale crop strategy. The detector training and evaluation stages are unchanged; all modifications are in the pseudo-label generation pipeline.

3.1. Baseline: Single-Scale MaskCut

The CutLER baseline follows a two-stage pipeline. In Stage 1 (pseudo-label generation), each training image is resized to 480×480 , processed by a frozen DINO ViT-S/8 backbone to extract key features $\mathbf{K} \in \mathbb{R}^{N \times d}$ for each of the N patches, and an affinity matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ is computed as $A_{ij} = \mathbf{k}_i^\top \mathbf{k}_j / \|\mathbf{k}_i\| \|\mathbf{k}_j\|$. The second-smallest eigenvector of the normalized graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is thresholded to produce a binary foreground mask, which is then refined with a Dense CRF. This process is repeated $K = 3$ times per image, masking out previously discovered objects before each iteration, producing up to three pseudo-masks per image. In Stage 2, Cascade Mask R-CNN with a ResNet-50 FPN backbone (initialized with DINO weights) is trained on these pseudo-labels class-agnostically.

3.2. DINO Feature-Contrast Heatmap

To identify image regions where the single full-image MaskCut pass is most likely to have missed objects, we compute a patch-level feature-contrast heatmap. For each patch i , we compute the mean cosine distance to its spatial neighbors:

$$h_i = 1 - \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \frac{\mathbf{k}_i^\top \mathbf{k}_j}{\|\mathbf{k}_i\| \|\mathbf{k}_j\|} \quad (1)$$

where $\mathcal{N}(i)$ is the set of patches adjacent to patch i in the spatial grid. High values of h_i indicate locally distinctive regions where the patch features differ substantially from their surroundings—these are candidate locations where small or locally salient objects may have been suppressed by global competition with more dominant structures.

The heatmap is upsampled to image resolution, smoothed with a Gaussian kernel, and thresholded at the p -th percentile (default $p = 85$) to select high-contrast seed locations.

3.3. Multi-Scale Crop Proposal and MaskCut Rescue

Around each high-contrast seed location, we place crop windows at three scales: $\{0.25, 0.35, 0.50\}$ of the original image dimensions. The crop center is placed at the seed, with boundary clamping to stay within the image. In addition, a spatial rescue grid adds crops at image corners and border midpoints regardless of heatmap values, ensuring coverage of objects near image boundaries. With a default

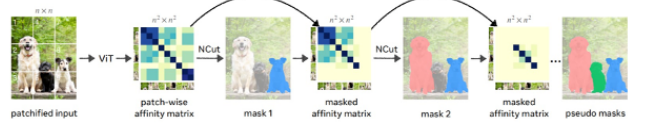


Figure 1. The MaskCut iterative segmentation procedure. Starting from the full image, MaskCut applies DINO feature extraction and Normalized Cuts iteratively, masking out found objects before each round.

budget of top- $k = 12$ heatmap seeds plus spatial rescue, a typical image receives approximately 15–20 crop windows per scale, for a total of 45–60 crop windows.

Each crop is resized to 480×480 and MaskCut is run independently, producing up to three masks per crop. Crop masks are back-projected to full-image coordinates using the inverse of the crop-to-full-image affine transform.

3.4. Filtering, Deduplication, and Merging

Crop-level masks undergo three filtering stages before merging with the baseline masks:

Fragment filtering: Masks with area below 0.5% of the image area or aspect ratio above 5:1 are discarded as likely edge or texture fragments.

Containment filtering: A crop mask that is more than 80% contained within an existing baseline mask is discarded (it is already covered).

Duplicate suppression: Among the remaining crop masks, pairs with $\text{IoU} > 0.5$ are deduplicated, keeping the higher-confidence mask (approximated by mask area stability across runs).

The surviving crop masks are then merged with the baseline single-scale MaskCut masks to produce the final combined pseudo-label set. The merge is additive: baseline masks are always kept, and only crop masks that pass all filtering stages are appended.

3.5. Pipeline Summary

The full pipeline for a single image is:

1. Run baseline full-image MaskCut to get masks $\mathcal{M}_{\text{base}}$.
2. Compute DINO feature-contrast heatmap $\mathbf{h}$.
3. Select top- k seed locations from $\mathbf{h}$ and add spatial rescue seeds.
4. For each seed and each scale $s \in \{0.25, 0.35, 0.50\}$, extract crop, run MaskCut, back-project.
5. Apply fragment, containment, and duplicate filtering to get rescue masks $\mathcal{M}_{\text{crop}}$.
6. Output $\mathcal{M}_{\text{base}} \cup \mathcal{M}_{\text{crop}}$ as the combined pseudo-label set.

4. Experiments

4.1. Datasets and Evaluation Protocol

Training data. We use a subset of TinyImageNet for pseudo-label generation and detector training. Our main 5-class study uses 2,500 images from five TinyImageNet classes. To validate scalability, we first ran a preliminary 10-class pilot on 500 images from ten classes. TinyImageNet provides a diverse set of natural images without CutLER-style class structure, making it a suitable domain-agnostic training source.

Evaluation. All detectors are evaluated class-agnostically on COCO val2017 using the standard MS-COCO AP metric suite. We report BBOX AP and SEGM AP as the primary metrics, along with size-stratified variants: APs (area $< 32^2$ px), APm (32^2 – 96^2 px), and API ($> 96^2$ px). We additionally conduct a direct pseudo-mask evaluation (Level 1) by comparing pseudo-labels generated on 500 COCO val2017 images against ground-truth instance masks using class-agnostic IoU matching at threshold 0.5.

4.2. Implementation Details

MaskCut configuration. All experiments use DINO ViT-S/8 with key features (`--vit-feat k`), patch size 8, and input resolution 480×480 . The number of masks per image is $K = 3$.

Hybrid pipeline configuration. The refined hybrid method uses crop scales $\{0.25, 0.35, 0.50\}$, heatmap percentile $p = 85$, and $\text{top-}k = 12$ heatmap seeds plus spatial rescue. These parameters were selected via a 100-image ablation study (described in Section 5).

Detector training. We train Cascade Mask R-CNN with ResNet-50 FPN backbone on the Bocconi HPC cluster using a single NVIDIA A100 80GB GPU. Training uses 20,000 iterations with batch size 8 and base learning rate 0.005. Copy-paste augmentation is enabled following the CutLER setup.

Runtime. Baseline single-scale pseudo-label generation takes approximately 4.5–5 hours for 2,500 images. The refined hybrid method requires approximately 16.5 hours (≈ 24 s/image), reflecting the cost of running MaskCut on 45–60 crops per image. The merge step takes only a few minutes.

4.3. Ablation Study Design

To identify the key hyperparameters of the hybrid pipeline without incurring the full 2,500-image generation cost, we ran an ablation study on a fixed 100-image subset. We varied the following factors independently:

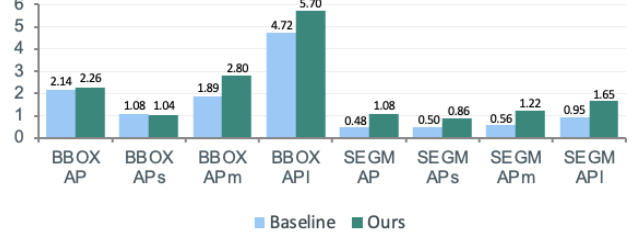


Figure 2. Baseline vs. Combined Multi-Scale detection results across all AP metrics (5 classes, 2500 images). The combined method improves overall BBOX AP and SEGM AP, with the largest gains in segmentation, while BBOX APs slightly decreases.

- **Heatmap percentile:** $p \in \{80, 85, 90\}$
- **Crop budget:** $\text{top-}k \in \{8, 12\}$
- **Crop scales:** default $\{0.25, 0.35, 0.50\}$ vs. tight $\{0.20, 0.30, 0.40\}$

We report generated mask count and qualitative inspection of mask diversity and noise as ablation criteria.

5. Results

5.1. Main Detection Results

Table 1 presents the main 5-class detector comparison on COCO class-agnostic evaluation. The baseline single-scale MaskCut achieves BBOX AP=2.14 and SEGM AP=0.48. The hybrid-only detector, trained on crop masks alone, performs much worse, with BBOX AP=0.30 and SEGM AP=0.19, confirming that the crop masks are too noisy to replace the original MaskCut masks. In contrast, the refined combined hybrid-best setting achieves BBOX AP=2.26 and SEGM AP=1.08, improving over the baseline in both overall detection and segmentation. The strongest gain is in segmentation, where AP improves by roughly $2.25 \times$ from 0.48 to 1.08. Small-object segmentation also improves, from SEGM APs=0.50 to 0.86, although small-object BBOX AP slightly decreases from 1.08 to 1.04.

5.2. Preliminary 10-Class Pilot

Prior to the main 5-class study, we ran a 10-class pilot on 500 images. The pattern was the same: hybrid-only (BBOX AP=0.11) was far below baseline (BBOX AP=2.22), while combined (BBOX AP=3.07) exceeded both. These pilot results confirmed that combining is the right design choice and informed the refined hybrid design used in the 5-class study.

5.3. Direct Pseudo-Mask Evaluation

Table 2 shows the direct evaluation of pseudo-masks against COCO ground truth on 500 validation images (3,799 GT objects: 1,570 small, 1,311 medium, 918 large). The key finding is that small-object recall remains almost zero for

Table 1. Main 5-class detector results on COCO class-agnostic evaluation (2,500 TinyImageNet training images).

Method	BBOX				SEGM			
	AP	AP _s	AP _m	AP _l	AP	AP _s	AP _m	AP _l
Baseline single-scale	2.14	1.08	1.89	4.72	0.48	0.50	0.56	0.95
Hybrid-only	0.30	0.41	0.25	0.92	0.19	0.25	0.22	0.63
Combined hybrid-best	2.26	1.04	2.80	5.70	1.08	0.86	1.22	1.65

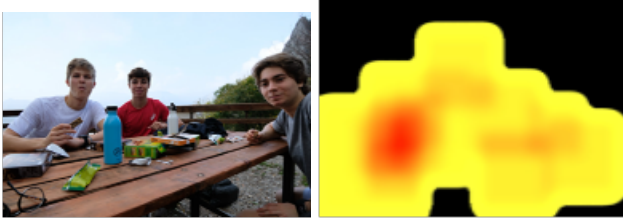


Figure 3. Left: example input image. Right: DINO feature-contrast heatmap, highlighting locally distinctive patches (red/yellow = high contrast, candidate crop seeds).



Figure 4. Additional pseudo-mask coverage from the combined method. Red and green masks are from the original CutLER MaskCut, while the blue mask comes from the crop-based pass. This qualitative case illustrates extra coverage, although direct COCO evaluation shows that small-object recall does not improve systematically.

both methods. Although crop rescaling increases the effective patch resolution of selected regions, this only helps when the crop contains a complete and well-localized small object. In practice, many heatmap-selected crops capture partial objects, textured background, or low-saliency regions. As a result, the generated crop masks often fail the IoU threshold against full COCO instances. This suggests that the bottleneck is not only patch resolution, but also crop proposal quality and mask completeness.

Multi-scale processing does improve medium-object recall by 18% (0.042→0.050) and large-object recall by 16% (0.376→0.435). These gains come from crop windows sampling different image regions, isolating medium and large objects that the single-pass global NCut missed.

5.4. Ablation Study

Table 3 shows the 100-image ablation results. The heatmap percentile ($p \in \{80, 85, 90\}$) had no effect on output mask count—all three settings produced identical results on this

Table 2. Direct pseudo-mask evaluation on 500 COCO val2017 images (class-agnostic, IoU ≥ 0.5).

Method	Small R	Med. R	Large R	Overall R
Baseline	0.0006	0.042	0.376	0.106
Multiscale	0.0006	0.050	0.435	0.122

Table 3. 100-image ablation study: effect of hyperparameters on final merged mask count.

Variant	Final merged anns	Notes
Baseline	496	reference
hp90	496	identical to baseline
hp80	496	identical to baseline
topk8	381	cleaner, fewer duplicates
tightcrop	816	noisier, more fragments

subset, suggesting the selected seed locations are robust to small percentile changes. Crop budget (top- k) had a clear effect: reducing from 12 to 8 seeds lowered final masks from 496 to 381, with qualitatively cleaner and less duplicated results. Crop scale had the largest effect: tight crops (0.20–0.40) increased final masks to 816 but also introduced more fragment noise. The default setting (0.25–0.50, top- $k=12$) was selected as a practical compromise between coverage, duplicate control, and fragment noise.

5.5. Qualitative Results

Figure 3 shows the DINO feature-contrast heatmap, which reliably highlights locally distinctive foreground regions. Figure 4 shows the improvement in pseudo-mask coverage when combining baseline and crop masks. The combined set captures multiple objects (persons, foreground objects) that the baseline pass missed. Figure 5 shows an isolated qualitative case where the combined detector predicts an additional small object, which indicates that the method might perform better with higher resolution images.

6. Conclusion

We presented a DINO feature-contrast heatmap-guided multi-scale extension of CutLER’s MaskCut pseudo-label generation that selectively crops and re-segments locally distinctive image regions, then merges the resulting masks with the original single-scale MaskCut output. Combin-

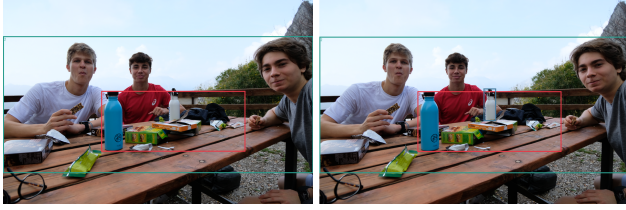


Figure 5. Recovered small object. Left: baseline detector. Right: combined multi-scale detector, recovering an additional small object (white water bottle).

ing multi-scale crop masks with the single-scale baseline achieves a $2.25\times$ improvement in segmentation AP ($0.48 \rightarrow 1.08$) and a modest gain in bounding-box AP ($2.14 \rightarrow 2.26$) on class-agnostic COCO evaluation. Direct pseudo-mask evaluation on 500 COCO validation images identifies the mechanism: crop windows sample different image regions, improving medium- and large-object recall by 18% and 16% respectively, rather than recovering small objects. Small-object recall remains near zero for both methods. In our experiments, heatmap-guided crop rescaling did not reliably produce complete, object-centered small-object masks. Although tighter crops can in principle increase patch coverage, many selected crops still led to partial or noisy NCut partitions. Multi-scale MaskCut therefore functions better as a pseudo-label coverage supplement than as a reliable small-object rescue stage. We note that the published CutLER result of AP=12.33 was obtained with multiple rounds of self-training on ~ 1.2 M ImageNet images; our experiments run Stage 1 and Stage 2 only, on 2,500 TinyImageNet images, so the absolute numbers are not directly comparable.

Limitations. The refined hybrid method is substantially more expensive than single-scale MaskCut (~ 24 vs. ~ 7 s/image), making it impractical for large-scale unsupervised training without further optimization. Our method’s masks alone are not enough to train a reliable detector. But, by combining with the baseline it increases the performance significantly. In our experiments, crop rescaling at fixed 480×480 input did not reliably overcome the small-object bottleneck: selected crops often still lacked enough object-specific DINO tokens for stable NCut separation. Addressing this limitation likely requires tighter object-centered crop proposals, higher-resolution inputs, smaller-patch or stronger backbones, or grouping methods less sensitive to sparse token support.

Future work. Three directions follow directly from these findings. First, increasing the number of patches per small object is the most direct path to crossing the spectral clustering stability threshold: raising input resolution from

480×480 to 640×640 with the same ViT-S/8 backbone increases small-object patch coverage by approximately $1.8\times$, and substituting DINOv2 [7]—which provides richer per-patch features and has demonstrated consistent gains on dense prediction tasks—offers a complementary improvement without changing input resolution. Second, the crop selector should be trained end-to-end rather than fixed to a heatmap threshold: our ablation showed that heatmap percentile had no effect on output mask count across the range $p \in \{80, 85, 90\}$, indicating that the fixed threshold is not an effective inductive bias for crop placement, and a learned selector could allocate the crop budget more efficiently. Third, the CutLER self-training stage was not applied in this study due to compute constraints; applying it to the combined pseudo-label set is a direct near-term extension that may further improve pseudo-label diversity across training rounds. An additional direction motivated by our training setup is to replace TinyImageNet (64×64 originals, heavily upsampled before pseudo-label generation) with full-resolution training images, which would reduce the domain gap with COCO evaluation and provide a cleaner test of the method’s ceiling.

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